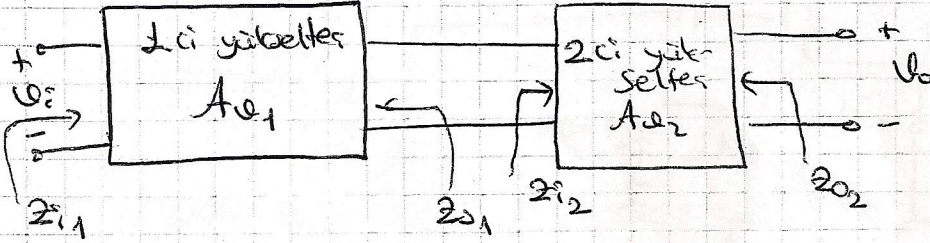


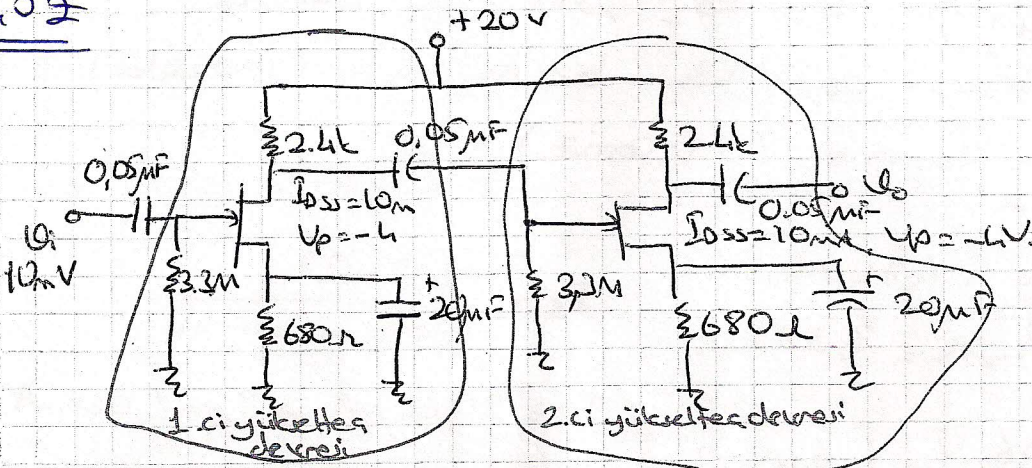
BÖLÜM-5:

Kaskad Yükleteçler:



$$A_e = A_{e1} \cdot A_{e2} \quad ; \quad Z_i = Z_{i1} \quad ; \quad Z_o = Z_{o2}$$

Örnek:
5.02



1.ci ve 2.ci yükleticinin DC analizi:

$$V_{GSQ} = -1,9V \quad ; \quad I_{DQ} = 2,8mA$$

$$g_{m0} = \frac{2I_{DSS}}{|V_p|} \left(1 - \frac{V_{GSQ}}{V_p} \right) = 2,6mS$$

$$A_{e1} = -g_m \cdot (R_D \parallel R_g) = -2,6m \cdot (2,4k \parallel 3,3M) = -6,23k$$

$$A_{e2} = -g_m \cdot R_D = -2,6m \cdot 2,4k = -6,24k$$

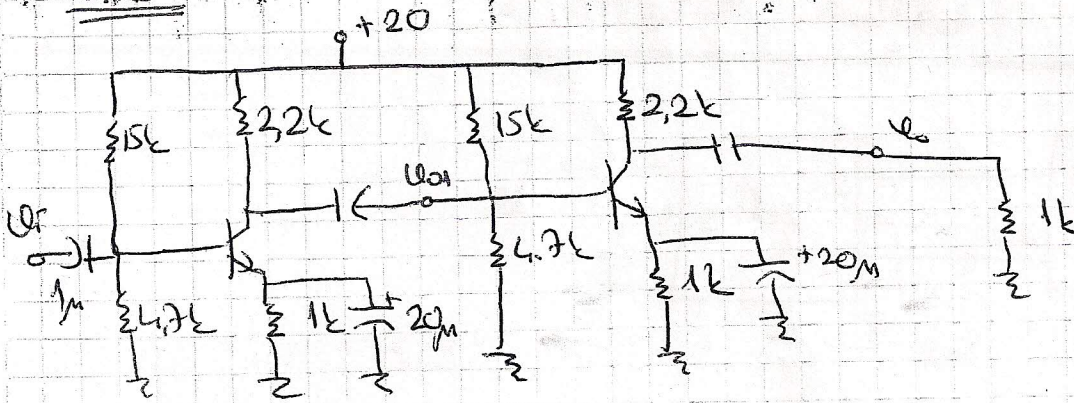
$$A_e = +38,87$$

$$U_i = 10mV \text{ de } U_o = 38,87 \times 10mV = 388,7mV/k$$

$$Z_i = Z_{i1} = 3,3M\Omega/k$$

$$Z_o = R_D = 2,4k\Omega/k$$

Örnek:



$\beta = 200$

a) DC analiz. $r_{e1}, r_{e2} = ?$

b) $A_{v1}, Z_{i1}, Z_{o1}, A_{v2}, Z_{i2}, Z_{o2}, A_{v0} = \frac{V_o}{V_i}, Z_i, Z_o = ?$

c) girme tepe değeri $25\mu V$ luk bir AC sinyal (sinüzoidal)

uygulanca V_{o1} ve V_o ne olur? (çizim.)

a) $V_{Th} = \frac{20}{15k + 4.7k}, 4.7k = 4.8V.$

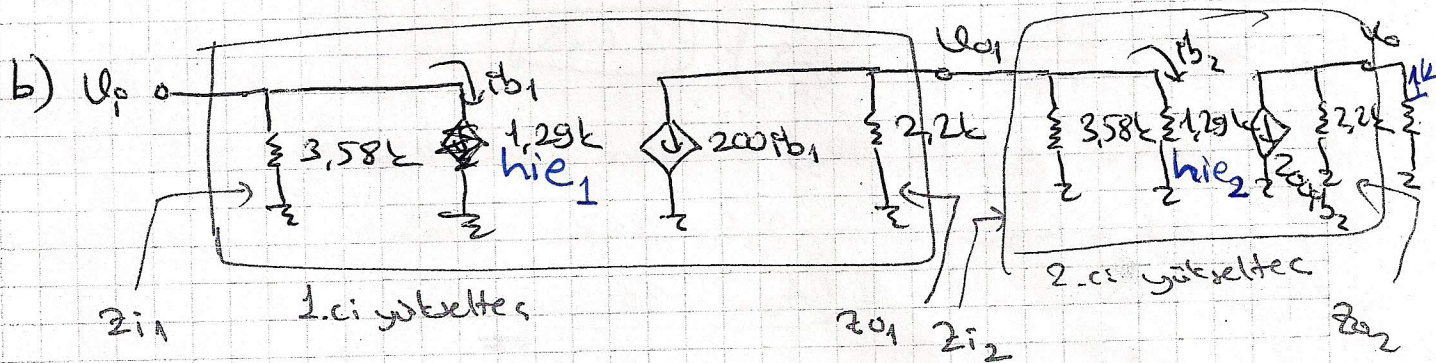
$R_{Th} = 15k \parallel 4.7k = 3.58k\Omega.$

$I_B = \frac{4.8 - 0.7}{3.58k + 201 \cdot 1k}$

$I_B = 20.06\mu A \Rightarrow I_E = 4.028mA.$

$r_{e1} = r_{e2} = \frac{26mV}{4.028mA} = 6.454\Omega$

$h_{re1} = h_{re2} = \beta r_{e1} = \beta r_{e2} = 1.29k\Omega$



2.ci çıkarttım girme direnci bir önceye yük olarak yazılır ve

$A_{v1} = -\frac{200}{1.29k} (2.2k \parallel Z_{i2})$

$Z_{i2} = 3.58k \parallel 1.29k$

$Z_{i2} = 0.95k$

$A_{v1} = -102.87$

$Z_{i1} = 0.95k, Z_{o1} = 2.2k$



$$A_{u2} = - \frac{200}{1,28k} (2,2k // 11k) = \underline{-106,6}$$

$$Z_{i2} = 0,95k \quad Z_{o2} = 2,2k$$

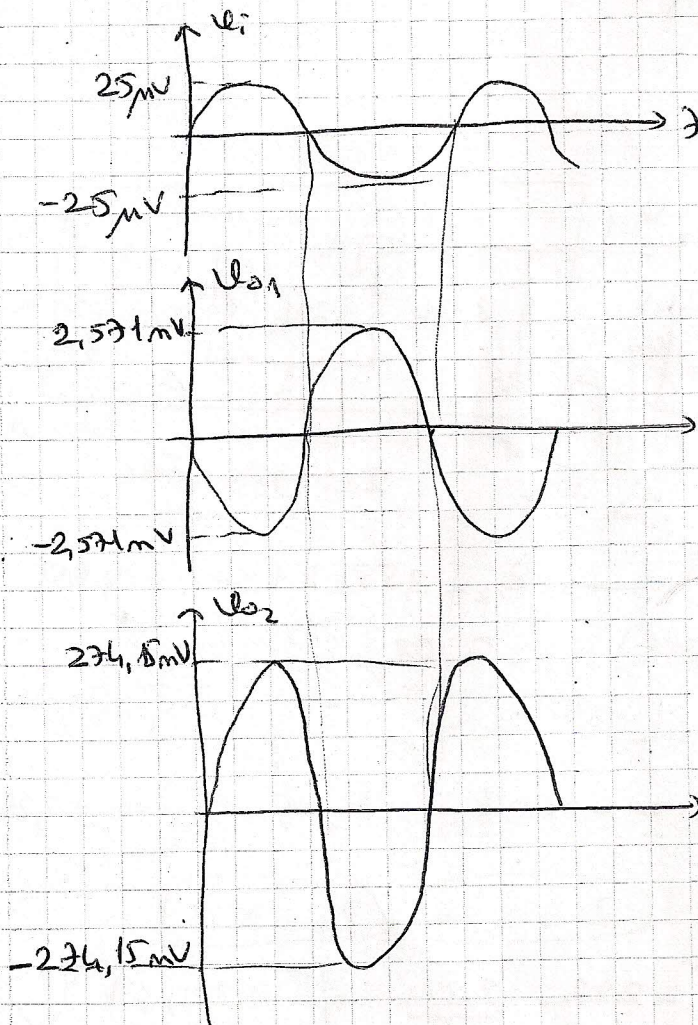
$$A_u = A_{u1} \cdot A_{u2} = \frac{(-102,87)(-106,6)}{10964,826} \quad Z_i = Z_{i1} = 0,95k$$

$$Z_o = Z_{o2} = 2,2k$$

c) $U_{o1} = U_i \cdot A_{u1} = 25\mu V \cdot (-102,87) = -2,571mV$

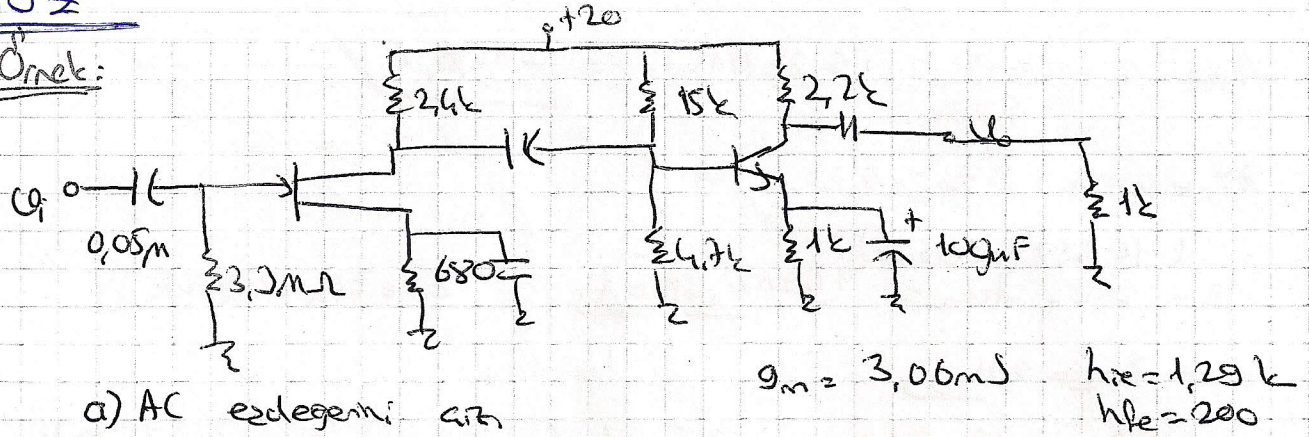
(- steht 180° für Polarisierung umgekehrt)

$$U_{o2} = A_{u2} \cdot U_{o1} = -106,6 \cdot -2,571mV = \underline{274,15mV}$$



GÖ2

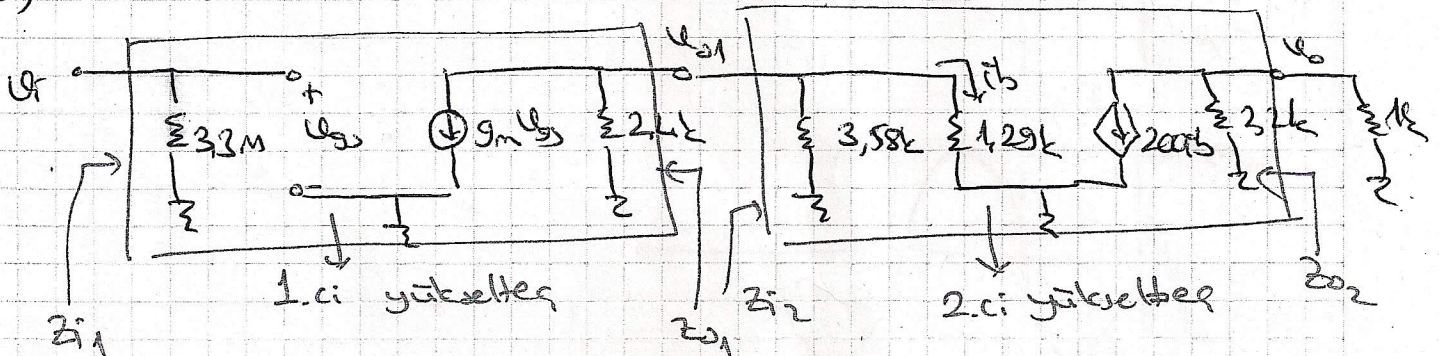
Örnek:



a) AC eşdeğeri çiz

b) A_{v1} , Z_{i1} , $Z_{o1} = ?$ A_v + $A_i = ?$ Z_i , $Z_o = ?$ A_{v2} , Z_{i2} , $Z_{o2} = ?$ c) $U_i = 10 \mu\text{V}$ de $U_o = ?$
 $I_i = 1 \mu\text{A}$ de $I_o = ?$

a)



$$b) A_{v1} = -g_m (2.4 \text{ k} \parallel Z_{i2}) \rightarrow Z_{i2} = 3.58 \text{ k} \parallel 1.29 \text{ k} = 0.95 \text{ k}$$

$$A_{v1} = -3.06 \cdot (2.4 \text{ k} \parallel 0.95 \text{ k}) = -2.08$$

$$Z_{i1} = 3.3 \text{ M} \quad , \quad Z_{o1} = 2.4 \text{ k}$$

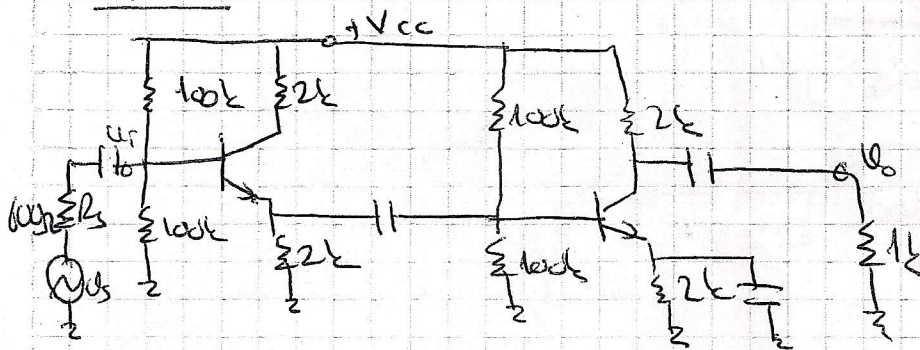
$$A_{v2} = -\frac{200}{1.29 \text{ k}} (2.2 \text{ k} \parallel 1 \text{ k}) = -106.6 \quad Z_{i2} = 0.95 \text{ k} \quad Z_{o2} = 2.2 \text{ k}$$

$$A_v = A_{v1} \cdot A_{v2} = \boxed{222} \quad \boxed{Z_i = 3.3 \text{ M}} \quad \boxed{Z_o = 2.2 \text{ k}}$$

$$A_i = -A_v \cdot \frac{Z_i}{R_y} = -222 \cdot \frac{3.3 \text{ M}}{1 \text{ k}} = \boxed{-732.6 \text{ k}} \quad \triangle$$

c) $U_i = 10 \mu\text{V}$ de $V_o = 2.22 \text{ mV}$. (faz farkı yok). $I_i = 1 \mu\text{A}$ de $I_o = -732.6 \text{ mA}$ (180° faz farkı olur.)

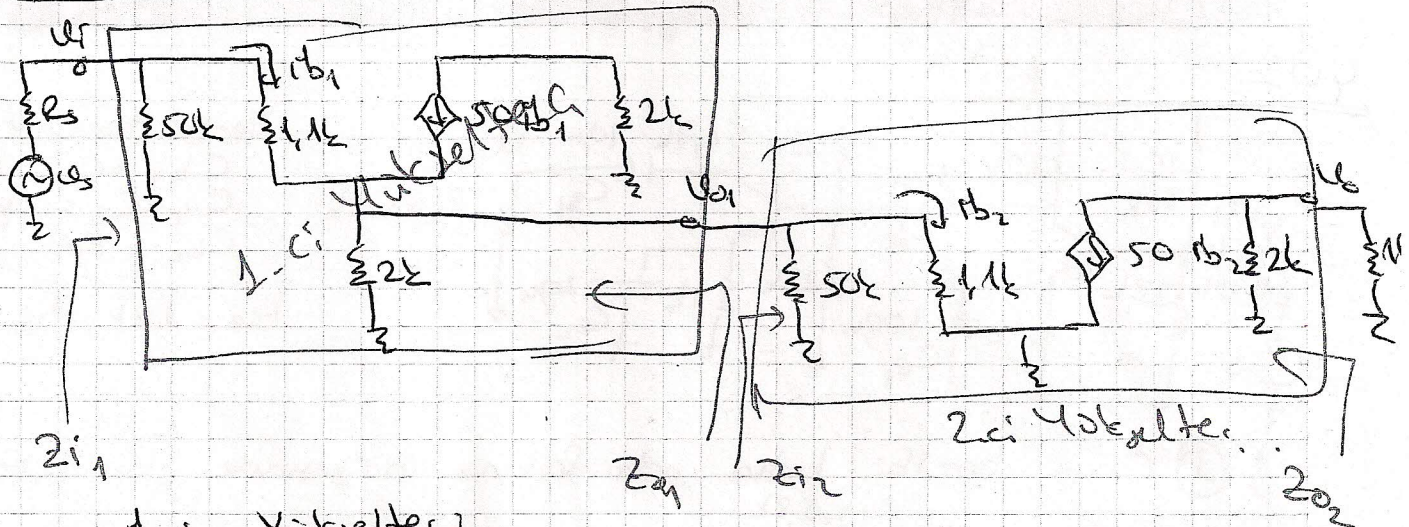
Örnek:



$$h_{re} = 1,1k$$

$$h_{fe} = 50$$

AC Edeğerleri:



1.ci Yükleliler:

$$Z_i = Z_{i1} = 50k \parallel Z_B \quad Z_B = u_i / i_{b1} \quad u_i = 1,1k i_{b1} + (2k \parallel Z_{i2}) \cdot 51 i_{b1}$$

$$Z_{i2} = 50k \parallel 1,1k = 1,076k \Rightarrow u_i = 53,98k i_{b1} \Rightarrow Z_B = 53,98k$$

$$Z_i = Z_{i1} = 50k \parallel 53,98k = 25,96k$$

$$A_{v1} = \frac{(2k \parallel Z_{i2}) \cdot 51 i_{b1}}{i_{b1} \cdot Z_B} = 0,98$$

$$Z_{o1} \text{ bulun } R_A = 0,6k \parallel 50k + 1,1k = 1,69k$$

$$Z_{o1} = 2k \parallel \frac{R_A}{\beta + 1} = 2000 \parallel 33,2 = 32,66k$$

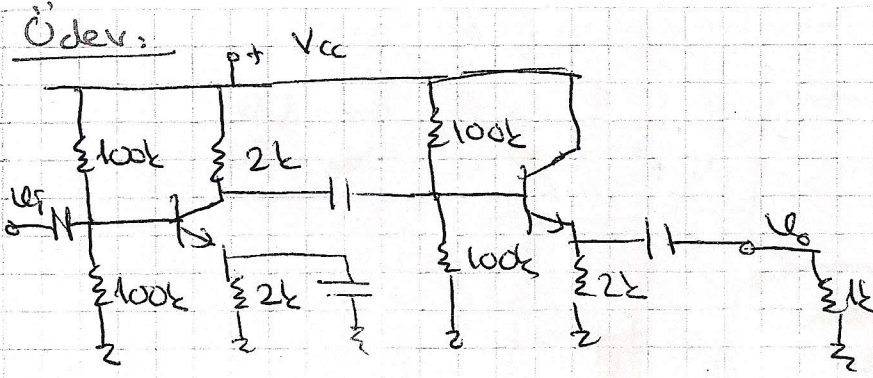
2.ci Yükleliler:

$$A_{v2} = -\frac{50}{1,1k} (2k \parallel 1k) = -30,3$$

$$Z_{i2} = 1,076k$$

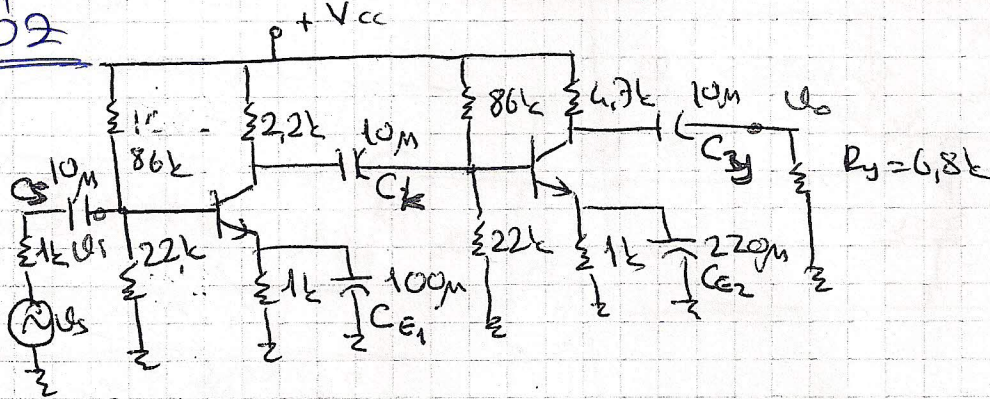
$$Z_{o2} = 2k$$

$$A_v = A_{v1} \times A_{v2} = -26,7$$



Kaskad Yükleteçlerde Frekans etkisi:

Ö2



Her iki transistör için
 $C_{be} = C_{ce} = 4pF$
 $C_{w1} = C_{w2} = 10pF$
 $C_{bc} = 8pF$

$h_{ie} = 1,2k$, $h_{fe} = 100$

① Önce her iki katın orta frekans bölgesinde kazançları hesaplanır:

$$Z_{i2} = (86k \parallel 22k) \parallel 1,2k = 1,123k$$

$$A_{v1} = -\frac{100}{1,2k} (1,123k \parallel 10M) = -\frac{0,943 \times 100}{1,2k} = \boxed{-61,95}$$

$$A_{v2} = -\frac{100}{1,2k} (4,7k \parallel 6,8k) = \boxed{-231,6}$$

② Sonra her iki katın giriş çıkış dirençleri hesaplanır:

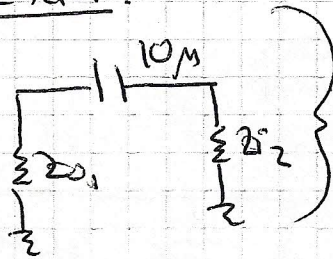
$$Z_{i1} = 1,123k \quad Z_{i2} = 1,123k \quad Z_{o1} = 22k \quad Z_{o2} = 4,7k$$

③ Alt kesim frekansı. (C_s , C_k , C_y , C_{e1} ve C_{e2} etkileri)

C_s için:

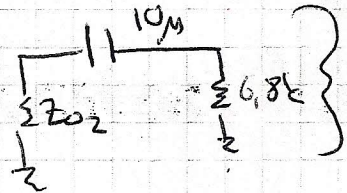
$$f_{C_s} = \frac{1}{2\pi \cdot 10 \cdot 10^{-6} \cdot (1k + 1,123k)} = 7,5 Hz$$

Ck sum:



$$f_{Ck} = \frac{1}{2\pi \cdot 10 \cdot 10^{-6} \cdot (1,123k + 2,2k)} = \underline{\underline{4,79Hz}}$$

Cy sum:



$$f_{Cy} = \frac{1}{2\pi \cdot 10 \cdot 10^{-6} \cdot (6,8k + 4,7k)} = \underline{\underline{1,38Hz}}$$

CE1 sum:

$$R_{eq1} = 1k \parallel \frac{R_{A1}}{101}$$

$$R_{A1} = (1k \parallel 17,52k) + 1,2k = 2,146k$$

$$R_{eq1} = 20,80\Omega$$

$$f_{CE1} = \frac{1}{2\pi \cdot 100 \cdot 10^{-6} \cdot 20,80} = \underline{\underline{76,55Hz}}$$

CE2 sum:

$$R_{eq2} = 1k \parallel \frac{R_{A2}}{101}$$

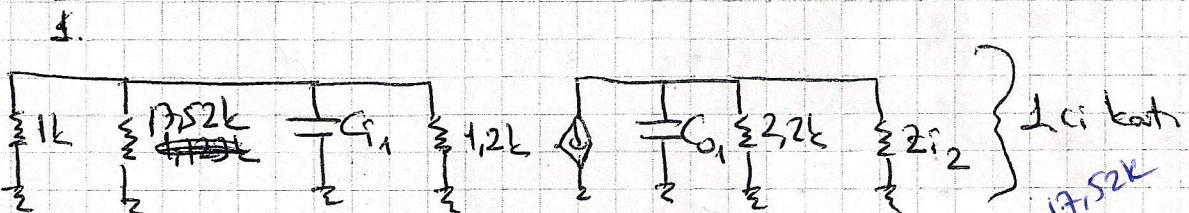
$$R_{A2} = (2,2k \parallel 17,52k) + 1,2k = 3,15k$$

$$R_{eq2} = 30,24\Omega$$

$$f_{CE2} = \frac{1}{2\pi \cdot 220 \cdot 10^{-6} \cdot 30,24} = \underline{\underline{23,93Hz}}$$

$$f_{dH} = 76,55Hz$$

④ Ust Kesim sum:



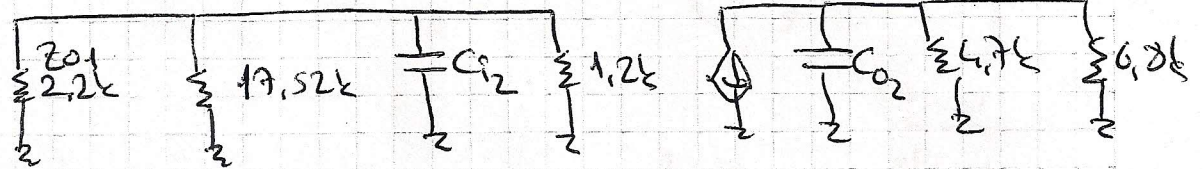
$$C_{p1} = C_{bc} (1 - A_{v1}) + 10p + 4p = 517pF, \quad 1k \parallel 1,123k \parallel 1,2k = 0,367k$$

$$f_{p1} = \frac{1}{2\pi \cdot 517 \cdot 10^{-12} \cdot 0,367k} = \underline{\underline{584kHz}}$$

$$C_{01} = \overbrace{C_{bc} \left(1 - \frac{1}{A_{v1}}\right)}^{\sim 8p} + 10p + 4p = 22pF. \quad 2,2k \parallel 17,52k = 0,74k$$

$$f_{C_{01}} = \frac{1}{2\pi \cdot 22 \cdot 10^{-12} \cdot 0,74k} = \underline{\underline{9,78 MHz}}$$

2. ci kat.:



$$C_{i2} = C_{bc} \left(1 - A_{v2}\right) + 10p + 4p = 1870pF.$$

$$2,2k \parallel 17,52k \parallel 1,2k = 0,74k$$

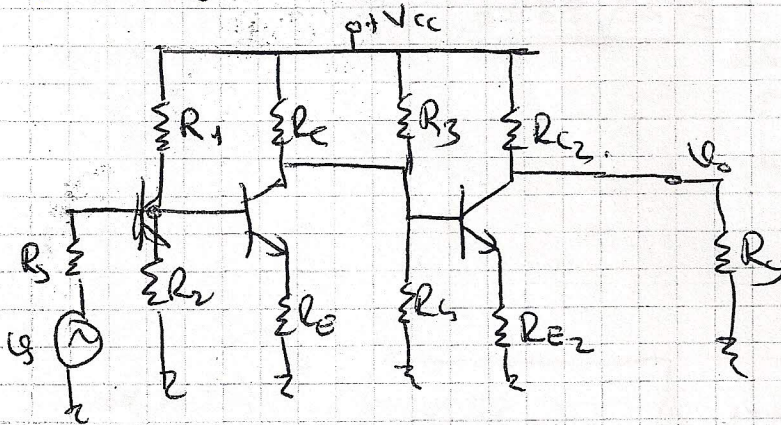
$$f_{C_{i2}} = \frac{1}{2\pi \cdot 1870 \cdot 10^{-12} \cdot 0,74k} = \underline{\underline{115 kHz}}$$

$$C_{02} = \overbrace{C_{bc} \left(1 - \frac{1}{A_{v2}}\right)}^{\sim 8p} + 10p + 4p = 22pF \quad 4,7k \parallel 6,8k = 2,8k$$

$$f_{C_{02}} = \frac{1}{2\pi \cdot 22 \cdot 10^{-12} \cdot 2,8k} = \underline{\underline{2,58 MHz}}$$

$$\boxed{f_{cut} = 115 kHz.}$$

X Doğrudan Bağlanmalı Kaskad Yitkileri:

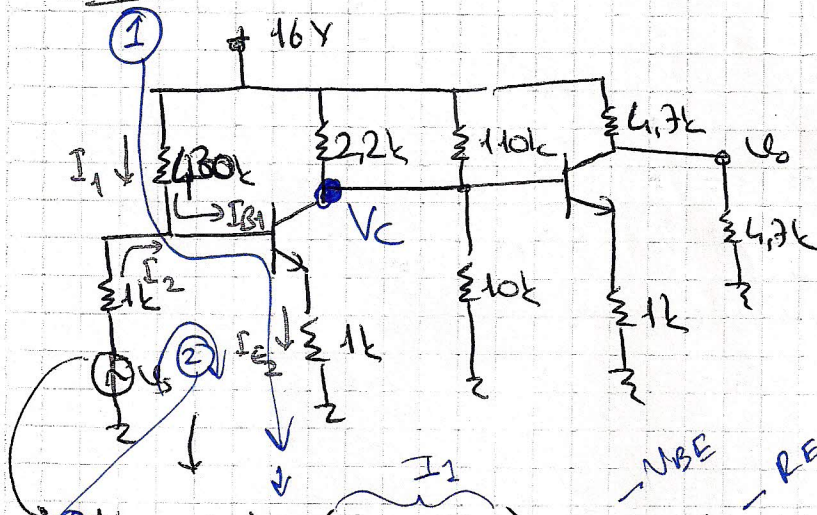


- AC analiz aynı
- DC analiz farklı.

Ümelk,

DC analyse $r_{e1}, r_{e2} = ?$

$\beta_1 = \beta_2 = 100$



① $16 = 430k \cdot (I_{B1} - I_2) + 0.7 + 1k \cdot 101 \cdot I_{B1}$

② $1k I_2 + 0.7 + 101 I_{B1} \cdot 1k = 0 \Rightarrow I_2 = \frac{-0.7 - 101k I_{B1}}{1k}$

(1) denkleminde I_2 yerine (2) denklemindeki I_2 ediyiz

$\Rightarrow 15.3 = 430k I_{B1} - 430k \cdot \left(\frac{-0.7 - 101k I_{B1}}{1k} \right) + 101k I_{B1}$

$\Rightarrow I_{B1} = 0.35 \mu A$

$I_{C1} = 35 \mu A \quad I_{E1} = 35.35 \mu A \Rightarrow r_{e1} = \frac{26m}{35.35 \mu A} = 735.2$

$\beta r_{e1} = 73.5k\Omega$

$V_C = 16 - 2.2k \cdot 35 \mu A = 15.923V$

$V_B = 15.923V$

$V_E = 15.232V$

$I_{E2} = 15.232 \mu A$ (Jant. kalay)

$V_{C1} = V_{B2} = \uparrow$

$V_B - V_{BE} = V_E$